

Expert Influence Is a Context-Conditioned Routing-Risk Field in Sparse Mixture-of-Experts Models

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Abstract

Sparse mixture-of-experts (MoE) models combine continuous expert computation with discrete top- k routing. We ask whether downstream expert influence is a stable global property or a local relation between a source expert, a context, and a router boundary. We intervene on post-gating expert contributions and define signed normalized margin erosion $\mathcal{R}_{i \rightarrow \ell, t}(x) = -\delta_i \partial_{\delta_i} \log \gamma_{\ell, t}$, where γ is the downstream top- k boundary margin. A derivative audit isolates an in-place forward-AD failure and validates an out-of-place amendment; a 512-prefix atlas then finds stable, concentrated marginal source sensitivity. However, a context-independent source-to-router graph fails its preregistered stability gate (edge Spearman 0.411; top-100 Jaccard 0.299). Signed local $\mathcal{R}$ is instead strongly informative: retrospective PR-AUC is 0.974, and two disjoint prospective studies show high-minus-matched-low increases of 35.9 and 26.6 percentage points in exact targeted transitions. A prospective component ablation then isolates the normalization: sensitivity-only and margin-only selectors each induce 0/64 targeted transitions, whereas maximizing $|g|/\gamma$ induces 32/64 versus 6/64 for a same-boundary matched control. Without maximizing over candidates, all 12 prospective transitions among 512 frozen near-boundary candidates lie in the highest risk quintile. Restoring changed expert IDs suppresses a median 99.7% of next-token KL across 225 observed transition records. Risk enrichment transfers across ten evaluated conditions in four architecture families, while checkpoint and rendered-context comparisons are heterogeneous. Moreover, at Base-anchored physical sites, the rank-boundary expert pair changes in 127/128 Qwen and 126/128 Kanana prefixes. Thus expert influence is a context-conditioned field whose local measurement transfers, but whose active boundary identity and operating geometry do not.

1 Introduction

Sparse MoE models scale capacity by sending each token to only a few experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Most MoE analyses focus on router probabilities, load balance, or activation counts. These describe which experts are selected, but not how changing an expert’s realized contribution changes the predictive distribution. That distinction matters when experts have redundant, synergistic, or opposing downstream effects.

We study a local functional geometry over expert contributions. At a chosen layer and token, we intervene after top- k selection and gate normalization, scaling expert i ’s realized contribution by $1 + \delta_i$. The derivative of a next-token log probability with respect to δ_i defines a functional score. Its second moment forms a positive semidefinite response matrix F , analogous to a Fisher matrix in expert-intervention space. Locally, $\frac{1}{2}\delta^\top F \delta$ predicts the KL divergence between baseline and intervened next-token distributions.

This local description coexists with a discontinuity. Routing at the intervened layer is fixed by construction, but the perturbed hidden state is propagated normally and later routers may change top- k sets. We therefore separate a smooth routing-stable regime from a routing-transition regime in which a perturbation crosses a boundary.

Our contributions are:

- We define a validated, architecture-aware routing-risk primitive, $\mathcal{R} = \epsilon|g|/\gamma$, for post-gating expert interventions and the score space actually ranked by each router.
- We show prospectively that both source susceptibility $|g|$ and boundary vulnerability γ^{-1} are necessary: neither component alone targets transitions, whereas their dimensionless product does. Risk also orders frozen, non-max-selected near-boundary candidates.
- We show that risk detection transfers across four MoE families, while field variation is heterogeneous and includes both boundary-identity turnover and changes in boundary geometry. Marginal sensitivity and static population graphs do not capture this conditional structure.

2 Related work

Sparse mixture-of-experts models. Sparsely gated layers increase capacity without proportional per-token computation (Shazeer et al., 2017). GShard and Switch Transformers developed scalable routing systems (Lepikhin et al., 2021; Fedus et al., 2022), and later work studied stable transfer at scale (Zoph et al., 2022). Routing frequency, however, does not measure an expert’s functional output effect.

Local geometry and interventions. Fisher information describes local second-order distributional change under parameter perturbations (Amari, 1998). Our coordinates are not model parameters; they are temporary dimensionless scalars on post-gating expert contributions. The discrete top- k router creates piecewise-smooth regions separated by routing boundaries, whose effect we measure explicitly.

Expert interpretation and routing boundaries. Prior work has examined expert specialization, routing frequency, and modular structure. Our question is narrower and causal-operational: given a realized source contribution, which downstream routing boundary can that contribution erode in the current context? We distinguish marginal importance from this directed, event-conditioned relation and test whether the latter survives prospective intervention.

3 Expert functional response

3.1 Contribution-space intervention

For one MoE layer and token, let $h \in \mathbb{R}^d$ be the layer input, $f_i(h)$ expert i ’s output, and $g_i(h)$ its normalized top- k gate. The realized contribution and baseline output are

$$u_i = g_i(h)f_i(h), \quad z_0 = b + \sum_{i=1}^E u_i. \quad (1)$$

We introduce $\delta \in \mathbb{R}^E$ and compute

$$z_\delta = b + \sum_{i=1}^E (1 + \delta_i)u_i, \quad \Delta z = U\delta. \quad (2)$$

The intervention is after gate normalization. It changes no expert input or parameter, performs no second normalization, and preserves the target layer’s baseline expert identities. Inactive experts have zero local score. Downstream transformer and router layers execute naturally on the perturbed hidden state.

3.2 Response matrix and KL prediction

For prefix x , let $p_0(y | x)$ and $p_\delta(y | x)$ be baseline and intervened next-token distributions. Define

$$s_i(x, y) = \frac{\partial \log p_\delta(y | x)}{\partial \delta_i} \Big|_{\delta=0}, \quad F = \mathbb{E}_{x, y \sim p_0(\cdot | x)} [s(x, y) s(x, y)^\top]. \quad (3)$$

One backward pass through differentiable δ yields all E coordinates. We accumulate outer products in FP64:

$$\hat{F} = \frac{1}{N} \sum_{n=1}^N s_n s_n^\top. \quad (4)$$

Under regularity within a fixed-routing region,

$$\mathbb{E}_x D_{\text{KL}}(p_0 \| p_\delta) = \frac{1}{2} \delta^\top F \delta + o(\|\delta\|_2^2). \quad (5)$$

We compare $\hat{D}_{\text{Full}} = \frac{1}{2} \delta^\top \hat{F} \delta$ with $\hat{D}_{\text{Diag}} = \frac{1}{2} \sum_i \hat{F}_{ii} \delta_i^2$.

3.3 Routing-transition diagnostic

For downstream layer ℓ and token t , let r denote the scalar score that the implementation actually ranks to select experts, and let $\gamma = r_{(k)} - r_{(k+1)}$ be the baseline top- k boundary gap. For modified scores r' , define

$$\rho_{\ell, t} = \frac{\|r'_{\ell, t} - r_{\ell, t}\|_\infty}{\gamma_{\ell, t}}, \quad \rho(x, \delta) = \max_{\ell, t} \rho_{\ell, t}. \quad (6)$$

A flip occurs when any monitored downstream top- k set changes. Because $\rho(x, \delta)$ is a maximum over a forward pass, it is a record-level risk summary, not an event-level calibrated probability or causal threshold.

3.4 Baseline-only signed router-margin risk

The observed displacement in Eq. equation 6 is useful diagnostically but is not available before intervention. We instead differentiate the baseline boundary margin. Thus the same r is used for both ρ and prospective risk; raw logits are not assumed to determine selection. Qwen uses its native top-8 ranking score; Nemotron and Kanana use $r_i = \sigma(q_i) + b_i$ with a rank-6/rank-7 gap; Gemma uses the FP32 softmax probability passed to top-8, before selected- weight renormalization. Contribution weights are treated separately from ranking scores. For source expert i and downstream event (ℓ, t) , let

$$g_{i \rightarrow \ell, t}(x) = \frac{\partial \gamma_{\ell, t}(x; \delta_i)}{\partial \delta_i} \Big|_{\delta_i=0}, \quad \mathcal{R}_{i \rightarrow \ell, t}(x; \delta_i) = -\delta_i \frac{g_{i \rightarrow \ell, t}(x)}{\gamma_{\ell, t}(x)}. \quad (7)$$

The sign of δ_i is chosen to erode the fixed baseline rank- k versus rank- $(k+1)$ margin. With magnitude ϵ , the resulting nonnegative score is $\mathcal{R} = \epsilon |g| / \gamma$. We also report the intrinsic normalized sensitivity $|g| / \gamma$, which removes the chosen intervention scale. A first order prediction crosses the boundary when $\mathcal{R} \geq 1$.

This ratio has a direct interpretation: $\epsilon |g|$ is the locally predicted perturbation-induced boundary displacement, while γ is the available routing safety buffer. A transition therefore requires two ingredients—source susceptibility and boundary vulnerability—whose dimensionless product is tested directly in the prospective component ablation. Equivalently,

$$\log \mathcal{R} = \log \epsilon + \log |g| - \log \gamma. \quad (8)$$

Checkpoint or context variation can therefore act through boundary-margin geometry, source-to-router derivative sensitivity, or both. The cross-family experiments test these channels separately.

This definition separates three objects. $D_{\ell, i} = \mathbb{E}[s_{\ell, i}^2]$ is a marginal source-output sensitivity; γ^{-1} is destination vulnerability; and $g_{i \rightarrow \ell, t}$ is the directed source-to-boundary relation. Only their signed, context-specific combination predicts whether this source perturbation consumes this boundary’s safety margin.

3.5 From a static graph to a local risk field

A natural hypothesis is to aggregate $-g/\gamma$ over prefixes into a global directed graph whose nodes are (ℓ, i) . We test this hypothesis rather than assume it. For source layer 23 we retain per-prefix signed source-to-downstream- layer edges and preregister split-half edge-rank, top-edge, sign, and out-strength stability. Failure motivates no post-hoc graph repair: the alternative scientific object is the full local field $\mathcal{R}_{i \rightarrow \ell, t}(x)$ evaluated anew for each context.

3.6 Prospective targeted intervention and routing clamp

For a new prefix, we compute $\mathcal{R}$ from the baseline forward only. The high condition maximizes it over eight active source experts and all downstream events. Holding its destination (ℓ^*, t^*) fixed, a matched-low source has at most half the high risk when possible and is matched on marginal sensitivity, activation frequency, global and local gate weight, and expert-output norm. A third active expert is selected by a deterministic random seed. The exact top- k change at (ℓ^*, t^*) is primary; any flip and next-token KL are secondary. Selection manifests are written and hashed before modified forwards.

Separately, for records already observed to flip, we restore baseline expert IDs at either the first changed event or all changed events while keeping every nonchanged event natural. This retrospective routing clamp asks how much observed divergence is removed by restoring expert choice. Conditioning on a flip prevents a population causal-effect interpretation.

4 Experimental protocol

Models and execution. The primary checkpoint is Qwen3-30B-A3B-Base. Cross-checkpoint experiments add Qwen3-30B-A3B and Qwen3-30B-A3B-Thinking-2507. These Qwen checkpoints have 48 MoE layers, hidden size 2,048, 128 experts, top-8 routing, and expert intermediate size 768. The Qwen source is zero-indexed layer 23 and destinations are layers 24–47. Cross-architecture experiments add NVIDIA Nemotron-3-Nano-30B-A3B-Base and its post-trained checkpoint. Nemotron has 52 hybrid Mamba/attention/MoE blocks, including 23 MoE blocks with 128 routed experts, top-6 routing, and one shared expert. Its source is physical block 27 (the 12th MoE block), and the 11 later MoE blocks are destinations; only routed contributions are perturbed. Gemma 4 experiments use 26B-A4B Base and IT weights. The executable model has 30 MoE text layers, 128 routed experts, top-8 routing, and no shared expert; its source is layer 14 and destinations are 15–29. We evaluate Base with the plain prompt and IT under the official thinking-OFF and thinking-ON chat renderings. Kanana-2 experiments use 30B-A3B Base and Thinking weights under the same plain prompt. Both have 48 decoder layers, with layer 0 dense and layers 1–47 MoE; each MoE layer contains 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. Qwen and Kanana use the same physical source layer 23 and destinations 24–47, matching normalized source depth while differing in attention, routing, and shared-expert structure. Expert parameters are layer-specific and IDs are not assumed semantically aligned across checkpoints or families. Models run in strict FP32, with TF32 disabled and FP64 KL evaluation. Seeds, configs, versions, scripts, selection hashes, and outcome hashes are stored.

Data and target protocol. We use GSM8K (Cobbe et al., 2021), pinned to revision 740312add88f781978c0658806c59bc2815b9866. A seeded permutation of the official train split produces 512 build and 128 disjoint held-out prefixes after filtering to 31–127 tokens. Answers are excluded from primary manifests. The prompt asks the model to solve the given problem and provide the answer. To avoid fixed-suffix routing collapse, the intervention is at the full-prompt token containing the question’s final non-punctuation content character; the evaluated distribution is the next token after the final prompt colon. The F build uses 512 prefixes; joint interventions use a fixed first 16-prefix held-out subset for the primary pilot. After that analysis was frozen, the same 40 $\epsilon = 0.01$ interventions were re-evaluated on all 128 held-out prefixes, producing 5,120 prefix records. This expanded run is a robustness and heterogeneity analysis, not an independent confirmation, and it does not access the sealed redesign confirmation split.

Two prospective G4-local splits contain 128 new prefixes each. The first is disjoint from all historical Stage 2 and reconstructed redesign IDs; the second also excludes the first G4-local split. Selection uses neither answers nor modified outputs. Cross-checkpoint and cross-architecture comparisons use the first prospective split, recomputing active experts, derivatives, destinations, and controls independently for every checkpoint/condition. Each uses its own 512-prefix marginal atlas. Gemma IT OFF and ON share weights but not input token sequences: OFF ends with an empty thought block, whereas ON adds a system turn containing `<|think|>`. We therefore treat them as two rendered contexts, not as an isolated causal intervention on thinking.

Response construction and interventions. For each build prefix, we sample $y \sim p_0(\cdot | x)$ with a fixed seed, compute $s(x, y)$, and accumulate $ss^\top$. The primary estimate activates 85 experts at least once; 52 are activated at least 10 times. Joint interventions sample only from these 52 coordinates. We cross $|S| \in \{1, 2, 4, 8\}$ with $\epsilon \in \{0.01, 0.03, 0.05\}$, drawing 10 interventions per cell (120 total). The primary local analysis is $\epsilon = 0.01$: 40 interventions and 640 intervention–prefix records. The full grid gives 1,920 records for routing analysis; the frozen-intervention expanded analysis gives 5,120 records across 128 prefixes.

Metrics and uncertainty. At intervention level we report R^2 , Pearson r , Spearman ρ_s , MSE, and MAE. We use 20,000 paired bootstrap resamples of the 40 local interventions for full-minus-diagonal comparisons. The expanded analysis reports intervention, prefix-cluster, and two-way bootstrap intervals, each with 20,000 replicates. Routing groups use one-sided Mann–Whitney tests. Mean log-KL contrast and overall flip probability use 20,000 prefix-cluster bootstrap replicates. Wilson intervals on binned routing curves are descriptive and do not account for clustering.

Protocol lineage and statistical units. Retrospective G2 uses 15,360 old held-out intervention–prefix records and is not independent confirmation. G3 uses 512 build prefixes and no modified forwards. Each prospective G4-local study uses 128 paired prefixes and three interventions per prefix; paired differences, 20,000-prefix bootstrap intervals, and exact McNemar tests are computed on prefixes. Cross-checkpoint effect differences use the same paired prefixes and are exploratory. Nemotron follows the same selection and outcome logic with a frozen family-specific numerical amendment: central finite differences with step 10^{-3} replace full-model forward AD, and margins are measured in the corrected routing-score space $\sigma(q) + b$. Gemma likewise uses central finite differences with step 10^{-3} and defines its margin in the exact pre-top- k FP32 softmax-probability space. Kanana also uses step- 10^{-3} central differences and the corrected $\sigma(q) + b$ rank-6/rank-7 score, while contribution weights use uncorrected sigmoid scores followed by top-6 normalization and scaling. Its shared MLP is never intervened upon. Multi-step FD audits pass for Gemma and Kanana, whereas Nemotron’s small-step full vectors are cancellation-limited and fail the same stability gate; we therefore treat Nemotron’s within-model risk enrichment as valid at the frozen oracle but do not infer a derivative shift mechanism. Routing clamps use all 225 existing flip-positive records and remain retrospective. Two additional prospective P0 studies use a fresh disjoint 128-prefix split: 64 prefixes for a five-condition component ablation and 64 for 512 non-max candidate interventions. Both selection manifests are frozen and hashed before modified forwards. No negative gate is overwritten, and the redesign confirmation file remains sealed.

5 Results

5.1 Marginal expert sensitivity is structured but not relational

Across 512 prefixes and all 48 Qwen layers we estimate $D_{\ell,i} = \mathbb{E}[s_{\ell,i}^2]$ together with activation frequency, mean gate weight, and expert input/output norms. Among 47 nondegenerate layers, the two 256-prefix halves have median within-layer Spearman 0.802 and top-16 Jaccard 0.600; even/odd splits give 0.791 and 0.684. Sensitivity is concentrated: median top-8 and top-16 shares are 0.877 and 0.958. Correlations with frequency, gate weight, output norm, and input norm are 0.834, 0.864, 0.845, and 0.824, but the top-8 Jaccard with conditional

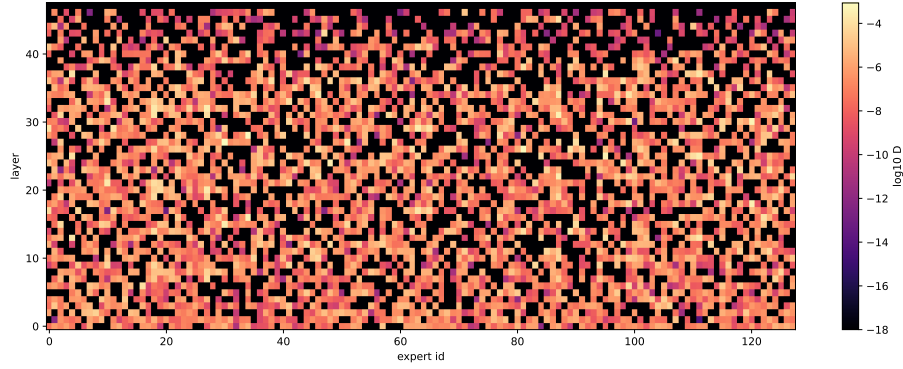


Figure 1: Source-output sensitivity atlas $D_{\ell,i}$ over 48 layers and 128 experts. Rankings are reproducible across prefix splits and highly concentrated, but this marginal atlas does not specify source-to-router relations.

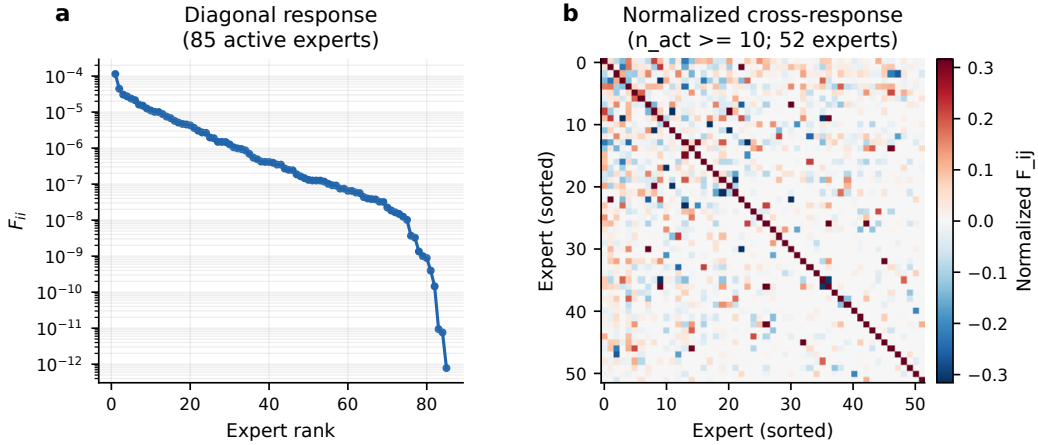


Figure 2: Estimated response geometry from 512 build prefixes. Left: F_{ii} for 85 experts activated at least once. Right: normalized signed cross-response for 52 experts activated at least 10 times. Coordinates outside observed routing support are not interpreted as zero functional response.

sensitivity is only 0.333 (0.143 at layer 23). Thus marginal expert sensitivity is reproducible and nonuniform, but it is a source attribute rather than a source-to-router relation.

Implementation validation. At $\delta = 0$, logits are bit-identical to baseline; inactive-expert scores and model-parameter gradient counts are zero; target routing is unchanged; repeated scores agree exactly. A formal 10-prefix finite-difference validation at layer 47 tests 80 active and 20 inactive cases at $\epsilon = 0.01$. Active-case median relative error is 0.735%, the 90th percentile is 4.942%, and signs agree in 100% of cases. Layer 47 minimizes accumulated downstream roundoff; the same intervention adapter and algebra are used at primary layer 23.

The $N = 512$ matrix is symmetric to machine precision, agrees with offline $S^T S/N$ to relative error 1.57×10^{-16} , and is positive semidefinite up to $\lambda_{\min} = -2.24 \times 10^{-20}$. Its trace is 4.17×10^{-4} , numerical rank is 80, and effective rank is 8.32. Figure 2 shows its diagonal and cross-response structure.

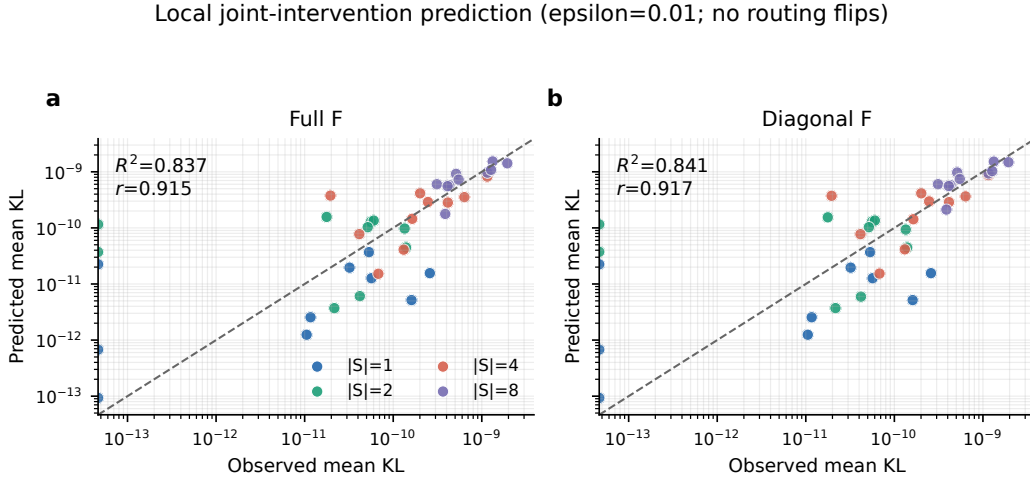


Figure 3: Local joint-intervention prediction at $\epsilon = 0.01$. Each point is observed or predicted next-token KL averaged over 16 held-out prefixes; colors indicate perturbed expert count. All 40 interventions preserve routing. Plotting zeros are placed below the smallest positive value only for log-scale display; statistics use unmodified values. Dashed line: identity.

Table 1: Local prediction at $\epsilon = 0.01$ (40 intervention-level means).

Predictor	R^2	Pearson r	Spearman ρ_s	MSE	MAE
Full F	0.837	0.915	0.783	3.239×10^{-20}	1.307×10^{-10}
Diagonal F	0.841	0.917	0.782	3.161×10^{-20}	1.305×10^{-10}

5.2 Global compression fails under prefix variation

The full response matrix, its off-diagonal structure, and a population source-to-router graph are three candidate global summaries. Each retains some signal, but none yields a stable relational object across prefixes.

Local full-matrix pilot. In the prespecified 16-prefix pilot, all 640 local prefix records preserve downstream routing. The full response achieves $R^2 = 0.837$, Pearson $r = 0.915$, and Spearman $\rho_s = 0.783$ (Figure 3, Table 1), supporting local quadratic prediction of unseen joint perturbations. The diagonal baseline is equally accurate: $R^2 = 0.841$, $r = 0.917$, and $\rho_s = 0.782$. A paired bootstrap gives $\Delta R^2 = -0.0040$ with 95% CI $[-0.0206, 0.0292]$ and bootstrap probability 0.379 that full R^2 exceeds diagonal. Thus we do not support the stronger hypothesis that estimated off-diagonal terms add predictive information.

For $|S| = 2, 4, 8$, median relative full-diagonal prediction differences are only 0.19%, 1.20%, and 3.14%. A matched high-interaction/control design appeared promising on 32 prefixes but failed to replicate on all 128 held-out prefixes. We do not retune groups on those outcomes or claim confirmation.

Expanded-prefix magnitude heterogeneity. We next reused the frozen 40 $\epsilon = 0.01$ interventions on all 128 held-out prefixes, without selecting new interventions or opening the sealed confirmation split. This produced 5,120 intervention-prefix records. The primary intervention-level means are substantially less favorable than the pilot: Full $R^2 = -0.0875$ and Pearson $r = 0.3016$, while Diagonal gives $R^2 = -0.0875$ and $r = 0.3078$ (Table 2). A two-way bootstrap with 20,000 replicates gives a 95% CI of $[-0.2140, -0.0236]$ for Full R^2 and $[-0.0024, -2.1 \times 10^{-7}]$ for Full minus Diagonal R^2 . The two fixed 64-prefix halves have Full Pearson correlations 0.1151 and 0.2748, exceeding the prespecified stability difference. Thus the expanded result is an engineering pass but a research robustness failure: local

Table 2: Expanded E4 local robustness using the frozen 40 interventions on 128 held-out prefixes. The all-record rows are primary; the no-flip rows are secondary diagnostics.

Prefix set	Predictor	n	R^2	Pearson r	Spearman ρ_s
All 128	Full	40	-0.0875	0.3016	0.8692
All 128	Diagonal	40	-0.0875	0.3078	0.8709
No-flip interventions	Full	28	0.4572	0.8529	0.8533
No-flip interventions	Diagonal	28	0.4360	0.8421	0.8533
Half A (64)	Full	40	-0.0291	0.1151	0.8750
Half B (64)	Full	40	-0.0585	0.2748	0.8432

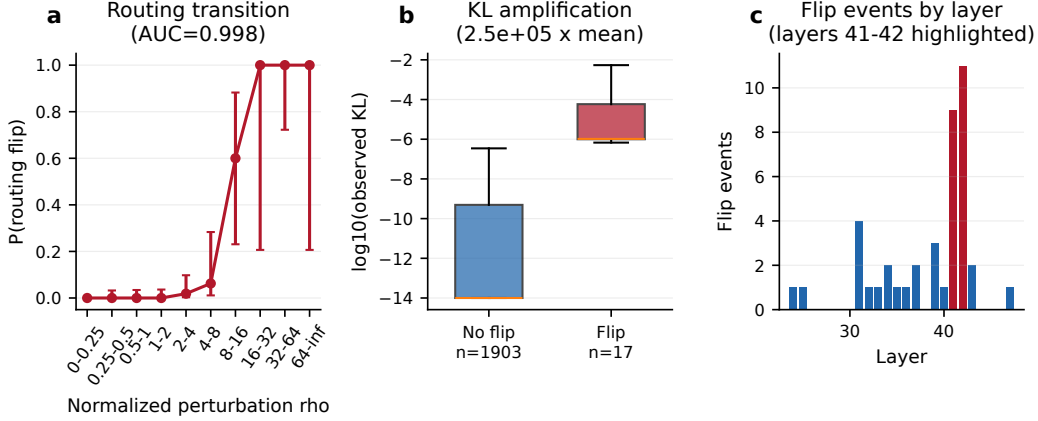


Figure 4: Routing transition and KL impact over 1,920 records. (a) Flip probability by prespecified bins of maximum normalized router displacement; bars show Wilson 95% intervals. (b) $\log_{10}(\text{KL} + 10^{-14})$ distributions with display outliers hidden. (c) Layers appearing in changed records; layers 41–42 are highlighted. These are record-level summaries.

prediction retains some ordering information in a routing-stable subset, but its absolute magnitude does not generalize reliably across prefixes.

The no-flip-intervention subset (28 of 40 interventions) has Full $R^2 = 0.4572$ and Pearson $r = 0.8529$, compared with Diagonal $R^2 = 0.4360$ and $r = 0.8421$. We report this only as a secondary heterogeneity diagnostic; removing flip-affected interventions is not a substitute for the all-record primary analysis. There are 20 routing-flip records (0.003906) in the expanded grid, and 12 of the 40 interventions are affected.

Observed routing-transition diagnostic. Across the full grid, 17 of 1,920 prefix records contain a downstream flip, comprising 41 layer/token changes and affecting 14 of 120 interventions. The record-level ρ statistic sharply identifies this transition (Figure 4a): ROC-AUC is 0.9977 and flip ρ is larger by a one-sided Mann-Whitney test ($p = 4.24 \times 10^{-16}$). No flip occurs among 1,833 records with $\rho < 2$; all 12 with $\rho \geq 16$ flip. These are descriptive ranges, not selected thresholds.

Mean KL is 6.82×10^{-4} for flip records and 2.76×10^{-9} for no-flip records, a ratio of 2.47×10^5 ; one-sided Mann-Whitney $p = 3.15 \times 10^{-16}$. A prefix-cluster bootstrap gives 95% CI [6.77, 9.62] for the difference in mean $\log_{10}(\text{KL} + 10^{-15})$. We emphasize distributions and the cluster interval because no-flip values contain many exact zeros and groups are highly imbalanced. Flips concentrate late: 9 and 11 changed-record occurrences involve layers 41 and 42.

Removing all 14 flip-affected interventions restores Pearson $r = 0.815$ for full F , but R^2 remains slightly negative because this broader no-flip set contains scales outside the validated

Table 3: Retrospective prediction of any routing transition (15,360 records).

Score	All PR-AUC	Half-B PR-AUC	Half-B ROC-AUC
Margin only	0.0277	0.0370	0.6990
D only	0.0717	0.0856	0.8326
Usage only	0.0727	0.0885	0.8306
D +margin	0.0809	0.1117	0.8652
Unsigned $\mathcal{R}$	0.6001	0.6081	0.9900
Signed $\mathcal{R}$	0.9736	0.9635	0.9953

local regime. No discrete flip is thus necessary but not sufficient for second-order truncation accuracy.

Estimator and spectrum instability. Nested estimates converge slowly: relative Frobenius errors to F_{512} are 0.856 at $N = 128$, 0.453 at $N = 256$, and 0.237 at $N = 384$. Two disjoint 256-prefix halves differ by 0.827, with top-8 subspace overlap 0.535. Four categorical token samples per prefix change the matrix by 0.353 relative to one sample, yet its two halves still differ by 0.762.

On a new redesign split, categorical Rao–Blackwell and dense-tail Hutchinson estimators improve probe convergence but leave prefix-half differences of 0.566–0.953. Build-only shrinkage selected $\alpha = 0.25$, but on an 11-intervention no-flip tuning smoke its MSE (4.177×10^{-19}) is slightly worse than raw (4.164×10^{-19}). We stopped this redesign and did not open the independent confirmation set (Appendix D).

Static relational graph. G3 aggregates per-prefix signed source-to-layer edges from 512 build prefixes. The two halves obtain edge Spearman 0.411 (gate 0.70), top-100 Jaccard 0.299 (gate 0.50), sign agreement 1.000, and source out-strength Spearman 0.387 (gate 0.70). Only direction passes. Residualizing D , usage, gate weight, output norm, and destination margin lowers edge correlation to 0.309. We retain this preregistered failure: above-null relational signal exists, but prefix variation prevents compression into a globally stationary graph.

5.3 Normalized local risk predicts routing transitions

We generate baseline features before joining labels for all 120 interventions and 128 held-out prefixes (15,360 records; 225 positives). Signed maximum $\mathcal{R}$ obtains PR-AUC 0.9736 overall and 0.9635 on the fixed second prefix half, compared with 0.0277 for margin only, 0.0717 for D only, 0.0727 for usage, and 0.0809 for D plus margin. Replacing the signed score with unsigned risk lowers PR-AUC to 0.6001, showing that direction matters.

The linear boundary-crossing rule gives 216 true positives, 3 false positives, 9 false negatives, and 15,132 true negatives: precision 0.986, recall 0.960, and specificity 0.9998. Exact predicted layer/token localization is 0.973 among flip records. Among 15,141 records with $\mathcal{R}_{\max} < 1$, nine flip (0.000594), below the frozen 0.005 safety tolerance but not zero. This analysis is retrospective because labels come from existing E5-A modified forwards; it motivates but does not substitute for prospective G4-local.

Prospective validation on two unseen splits. G4-local tests the local field directly. For each new prefix, all selection features are baseline-only; the high, matched-low, and random conditions share the exact destination event and margin. The selection manifest is hashed before modified forwards. No matched-low selection is relaxed.

On the first independent split, high-risk sources flip the targeted event in 59/128 prefixes (0.4609), versus 13/128 (0.1016) for matched-low and 15/128 (0.1172) for random. The primary paired high-minus-low effect is 0.3594 with 95% prefix-bootstrap CI [0.2812, 0.4453]; all 46 discordant pairs are high-only (exact McNemar $p = 2.84 \times 10^{-14}$). On a second disjoint split, rates are 46/128 (0.3594), 12/128 (0.0938), and 16/128 (0.1250), with primary effect 0.2656 [0.1875, 0.3438] and 34 high-only discordant pairs ($p = 1.16 \times 10^{-10}$). Both indepen-

Table 4: Prospective exact targeted-event routing transitions.

Study	High	Matched low	Random	High-low (95% CI)
G4-local 1	0.4609	0.1016	0.1172	0.3594 [0.2812,0.4453]
G4-local 2	0.3594	0.0938	0.1250	0.2656 [0.1875,0.3438]

Table 5: Prospective component ablation on 64 frozen prefixes. Medians are computed over one selected intervention per prefix; intervals are prefix-bootstrap 95% CIs for the exact target-flip rate.

Selector	Median $ g $	Median γ	Median $\mathcal{R}$	Target flip
Sensitivity-only	.0321	.4424	.00215	0/64 (.000)
Margin-only	.000910	.00608	.00457	0/64 (.000)
Normalized risk	.0778	.00234	.970	32/64 (.500 [.375,.625])
Matched control	.0149	.00234	.143	6/64 (.094 [.031,.172])

dently frozen gates pass. The secondary fixed-effect summary is 0.3086 [0.2523,0.3649], but $I^2 = 62.2\%$; we therefore emphasize study-specific estimates.

These are prospective targeted validations, not randomized treatment effects: source conditions are selected from baseline geometry. Nevertheless, matching fixes the destination and balances source marginals, selection precedes outcomes, and the exact preselected event is primary. The evidence therefore establishes predictive intervention utility of the relational score, rather than merely post-intervention association.

Prospective component ablation. On a fresh disjoint 64-prefix split, we froze five baseline-only selectors before modified forwards. The sensitivity-only selector chooses large $|g|$ together with a large available buffer; the margin-only selector chooses a small γ and a moderate derivative. Both produce 0/64 exact target flips (Table 5). In contrast, maximizing $|g|/\gamma$ gives 32/64 exact flips (0.500, prefix-bootstrap 95% CI [0.375,0.625]), versus 6/64 (0.094 [0.031,0.172]) for a source-matched control at the same destination boundary. The large-derivative condition has more than 35 times the median $|g|$ of the low-margin condition, yet both have small normalized risk and neither crosses its boundary. Thus routing transition requires both source susceptibility and boundary vulnerability; $\mathcal{R}$ is not a repackaging of either component alone.

Prospective ordering without max-selection. The other 64 prefixes in the fresh split test the field away from the targeting maximum. For each prefix, we deterministically sample one event uniformly from its 16 smallest-margin downstream events, retain all eight active source directions, freeze the resulting 512 candidates, and only then run modified forwards. Exact target-flip rates over global frozen-risk quintiles are [0,0,0,0,0.118]; all 12 positives occur in the highest quintile, giving PR-AUC and ROC-AUC 1.0 on this split. Because positives remain sparse and the top-quintile prefix-cluster interval is [0.024,0.219], this establishes prospective ordering conditional on near-boundary candidate events, not a globally calibrated flip probability. Together with retrospective localization and max-risk targeting, it closes the intended association-targeting-component necessity-prospective ordering chain.

5.4 Routing restoration accounts for downstream divergence

We next analyze all 225 old E5-A records with an observed transition. Restoring all changed expert IDs suppresses a median 0.99677 of natural next-token KL; 96.9% of records suppress more than half and 88.4% suppress more than 90%. Restoring IDs and baseline weights yields a similar median (0.99661). Restoring only the first transition has the same median suppression; its KL equals the all-flip clamp exactly in 216/225 cases. Nine exceptions, including two negative first-clamp restoration fractions, show that later transitions can interact nonadditively. Zero-intervention clamps are exactly inert and natural flip counts

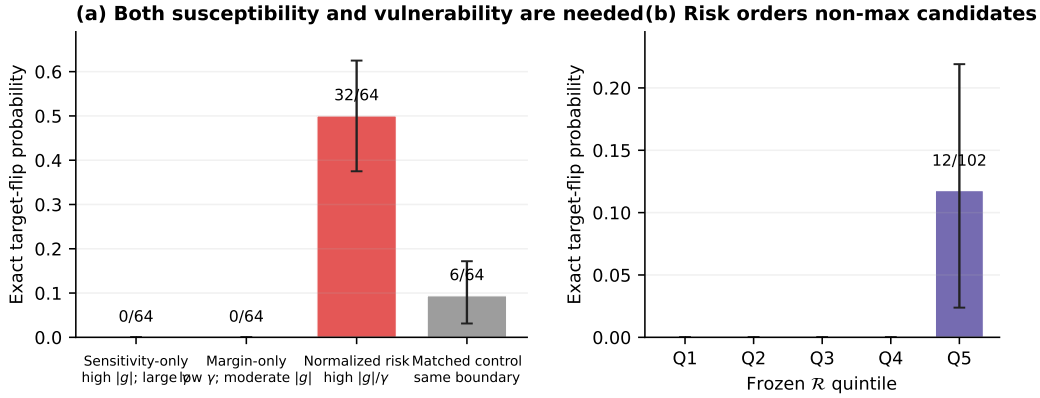


Figure 5: Prospective evidence that normalized risk is necessary and not an artifact of maximum selection. (a) Sensitivity-only and margin-only selectors do not induce their targeted transition; $|g|/\gamma$ does and exceeds a same-boundary matched control. (b) For 512 candidates frozen without maximizing risk, all observed transitions fall in the highest risk quintile. Bars show prefix-cluster bootstrap 95% intervals.

Table 6: Prospective G4-local outcomes in nine weight states and ten evaluated conditions. Rates are within-condition; cross-family levels are not directly comparable because architecture-specific top- k , source depth, and derivative oracles differ.

Family	Checkpoint	High	Low	Random	High-low (95% CI)
Qwen3	Base	.4609	.1016	.1172	.3594 [.2812,.4453]
Qwen3	A3B	.4375	.1016	.1250	.3359 [.2500,.4219]
Qwen3	Thinking-2507	.2734	.0547	.0703	.2188 [.1484,.2891]
Nemotron	Base	.2344	.0625	.0859	.1719 [.1094,.2422]
Nemotron	Post	.2188	.0703	.0781	.1484 [.0859,.2188]
Gemma 4	Base	.1484	.0234	.0391	.1250 [.0703,.1875]
Gemma 4	IT-OFF	.3906	.1406	.1406	.2500 [.1797,.3281]
Gemma 4	IT-ON	.2578	.0547	.0625	.2031 [.1328,.2812]
Kanana-2	Base	.5859	.1719	.1797	.4141 [.3203,.5078]
Kanana-2	Thinking	.4141	.1250	.1172	.2891 [.2109,.3672]

are reproduced in 225/225 cases. Routing restoration therefore accounts for most divergence in observed transition records. Because the analysis conditions on flips and is retrospective, it is not a population causal-effect estimate.

5.5 Risk detection transfers; field variation is heterogeneous

We next separate two questions: whether local risk detects transitions within a model, and whether weight state or rendered context changes its distribution. We rerun G4-local on Qwen3-30B-A3B, Qwen3-30B-A3B-Thinking-2507, a distinct Nemotron Nano Base/Post pair, Gemma 4 Base, IT-OFF, and IT-ON, and Kanana-2 Base/Thinking using the same 128 prefixes. Every evaluated condition receives its own 512-prefix marginal atlas and independently recomputed active experts, derivatives, destinations, and controls. Selections are frozen before outcomes. All ten within-condition gates pass (Table 6).

A3B minus Qwen Base changes the paired high-low effect by -0.0234 ($[-0.1406, 0.0938]$), resolving no difference. Thinking minus Base is -0.1406 ($[-0.2500, -0.0312]$): Thinking has lower absolute random susceptibility while retaining $3.89\times$ high/random and $5.00\times$ high/low enrichment. The Nemotron Post-minus-Base effect is -0.0234 ($[-0.1172, 0.0703]$), also unresolved; random rates are similar (.0781 versus .0859). Yet Nemotron high/random enrichment is $2.73\times$ for Base and $2.80\times$ for Post. Local risk thus remains predictive even when no checkpoint shift is detected. Gemma also passes in Base, IT-OFF, and IT-ON;

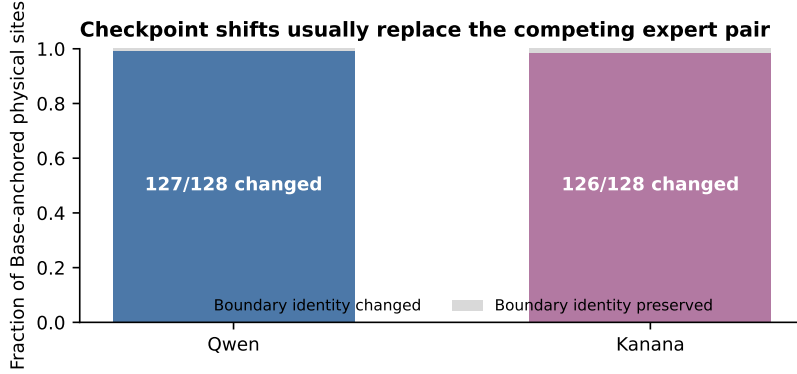


Figure 6: Boundary-identity turnover at Base-anchored physical router sites. The rank- k /rank- $(k+1)$ expert pair changes in 127/128 Qwen Base-Thinking and 126/128 Kanana Base-Thinking comparisons. A common physical layer/token therefore does not generally define a common routing boundary across checkpoints.

Table 7: Within-family checkpoint contrasts at frozen selected events. Effect is the paired change in high-minus-low flip rate; geometry entries are paired geometric-mean ratios with 95% prefix-bootstrap intervals.

Pair	Effect difference	γ ratio	$ g $ ratio	$ g /\gamma$ ratio
Qwen Think./Base	-.1406 [-.2500,-.0312]	1.57 [1.07,2.29]	.80 [.61,1.04]	.51 [.38,.69]
Nemotron Post/Base	-.0234 [-.1172,.0703]	1.33 [.84,2.02]	1.19 [.89,1.57]	.89 [.61,1.35]
Gemma IT-ON/OFF	-.0469 [-.1484,.0547]	2.02 [1.29,3.13]	1.21 [.86,1.73]	.60 [.44,.82]
Kanana Think./Base	-.1250 [-.2422,0]	1.02 [.64,1.62]	.66 [.45,.95]	.64 [.44,.94]

high/low enrichment is $6.33\times$, $2.78\times$, and $4.71\times$, respectively. Kanana Base and Thinking also pass, with high/low enrichment $3.41\times$ and $3.31\times$. Their ascending-risk flip rates are sharply stratified: $[0, .038, 1, 1, .92]$ and $[0, 0, .231, .92, .96]$.

Table 7 compares each family’s checkpoint pair at its independently selected high-risk events. For Qwen, the decomposition $\mathcal{R} = \epsilon|g|/\gamma$ associates Thinking’s lower normalized risk primarily with the denominator: the paired geometric-mean margin ratio is 1.57 [1.07,2.29], while $|g|$ is 0.80 [0.61,1.04], yielding $|g|/\gamma = 0.51$ [0.38,0.69]. Extending the derivative summary to all eight active sources at each selected event leaves derivative shrinkage unresolved. Ordinary A3B differs from Base on neither component. Nemotron shows no resolved geometry shift: all three intervals include one. At fixed Gemma IT weights, ON/OFF margin is 2.02 [1.29,3.13] and $|g|/\gamma$ is 0.60 [0.44,0.82], but its high-low effect change is -0.0469 $([-0.1484,0.0547])$ and is unresolved. Kanana Thinking/Base instead has stable selected margins (1.02 [.64,1.62]) but lower $|g|$ (.66 [.45,.95]) and $|g|/\gamma$ (.64 [.44,.94]), providing a derivative-associated counterexample to a universal margin account. Its high-low effect changes by -0.1250 $([-0.2422,0])$; the interval touches zero, so the discrete checkpoint contrast remains unresolved under the strict criterion.

A Base-anchored common-site audit removes layer/token reselection but reveals a second, more basic variation channel (Figure 6). At the same physical (ℓ, t) , the exact rank-boundary expert pair is preserved in only 1/128 Qwen and 2/128 Kanana prefixes. Thus checkpoint variation changes not only the margin and derivative of a local competition, but usually which experts are competing. State-specific margin ratios at these sites are consequently not same-boundary effects; we report them only in the appendix. The main conclusion is boundary-identity turnover, which also helps explain why a stationary population graph loses relational edge identity.

High-condition median next-token KL is 1.79×10^{-7} for Qwen Base, 1.97×10^{-7} for A3B, and 1.48×10^{-8} for Thinking. Nemotron’s corresponding medians are 1.80×10^{-8} for Base

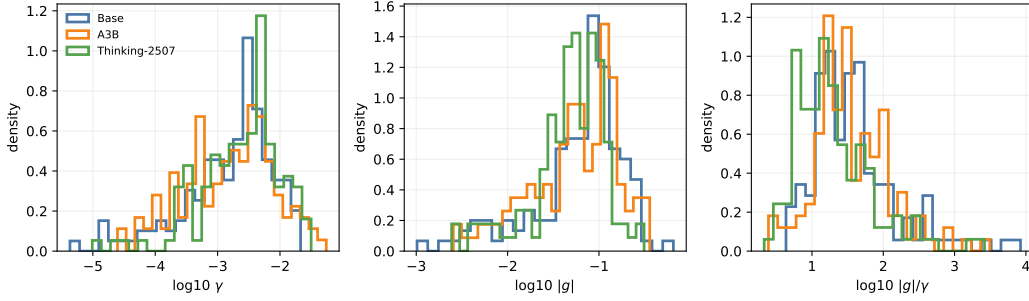


Figure 7: Distribution of selected-event router margin γ , derivative magnitude $|g|$, and intrinsic normalized sensitivity $|g|/\gamma$ across checkpoints. Thinking-2507 shifts normalized erosion downward primarily through larger selected-event margins.

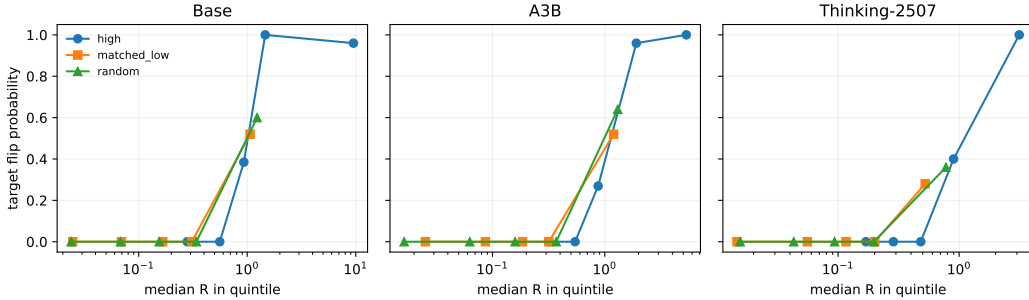


Figure 8: Observed exact target-flip probability over ascending within-model quintiles of frozen baseline risk. Thinking shifts the risk distribution left without eliminating its monotone predictive structure. Quintiles are diagnostic and are not a shared calibrated probability scale.

and 3.49×10^{-9} for Post, despite no resolved flip-susceptibility difference; continuous output sensitivity and discrete transition susceptibility are therefore related but not identical. No checkpoint has a next-token top-1 change. We interpret Thinking-2507 as exhibiting lower prompt-boundary local routing susceptibility, not as evidence that generated chain-of-thought improves final-answer accuracy. Only 26.6% (A3B) and 28.1% (Thinking) of high-source IDs match Qwen Base, and exact target-event agreement is 3.1% and 4.7%, reinforcing checkpoint conditioning.

Gemma sharpens this distinction. Base, IT-OFF, and IT-ON all pass prospectively, with high-low effects .1250, .2500, and .2031. IT-OFF versus Base has smaller $|g|$ (ratio .63) but a much smaller margin (ratio .23), so normalized risk rises $2.73\times$: raw derivative magnitude alone is not susceptibility. Holding IT weights fixed, thinking ON roughly doubles the selected margin and reduces $|g|/\gamma$ to .60 of OFF, although the observed high-low effect change remains unresolved. This directly supports context conditioning, but not a causal claim about internal thinking, because the official mode switch changes the rendered input and prediction context.

Kanana provides an independent architectural validation and a qualitatively different associated mechanism. Under the same plain prompt and matched physical source depth as Qwen, Base and Thinking high-low effects are .4141 and .2891. The paired difference is directionally negative but unresolved because its 95% interval ends at zero. Continuous selected geometry is resolved: γ remains unchanged while $|g|$ and $|g|/\gamma$ fall to .66 and .64 of Base. Thus local risk is not a repackaged margin statistic; source-to-router susceptibility can change at the same available routing buffer. Median high-condition KL also falls from

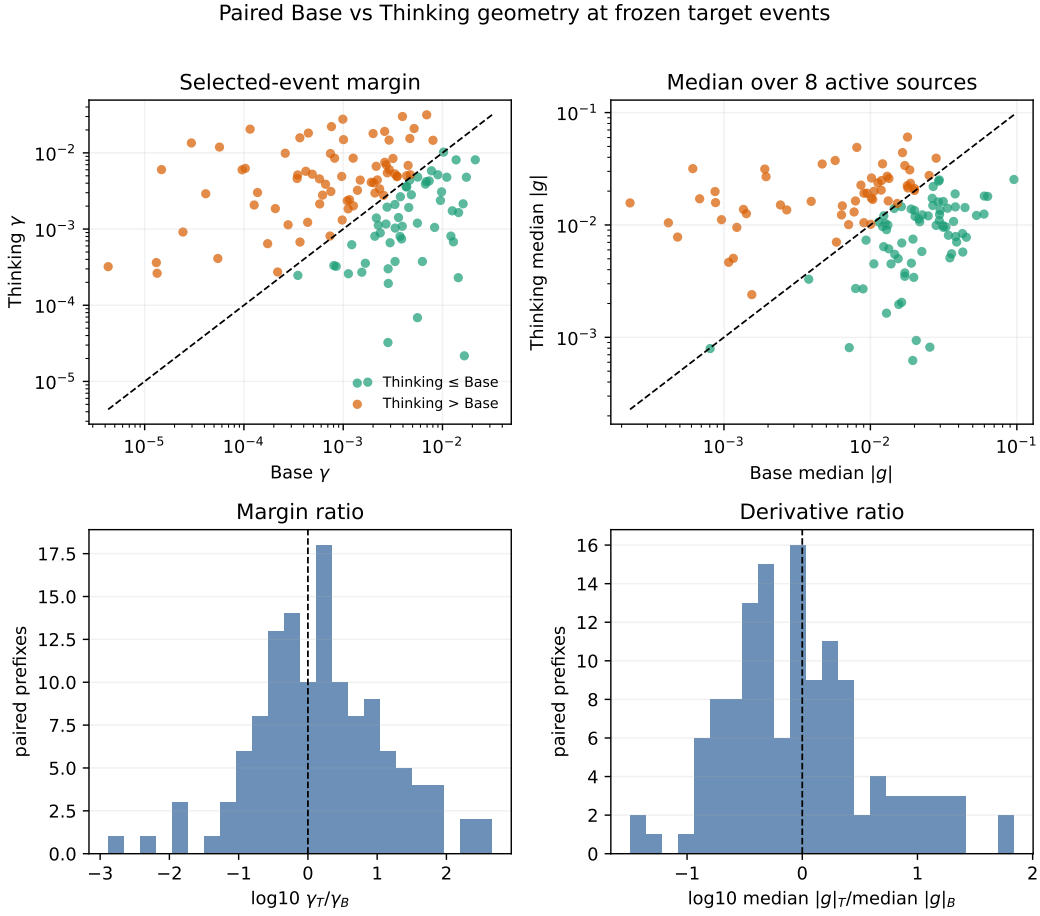


Figure 9: Paired Base–Thinking geometry for 128 prefixes at each checkpoint’s independently selected target event. Thinking has larger γ in 73/128 pairs and smaller eight-source median $|g|$ in 72/128. Aggregate margin expansion coexists with substantial context heterogeneity. These selected events do not represent the unmeasured unconditional distribution of all eligible events.

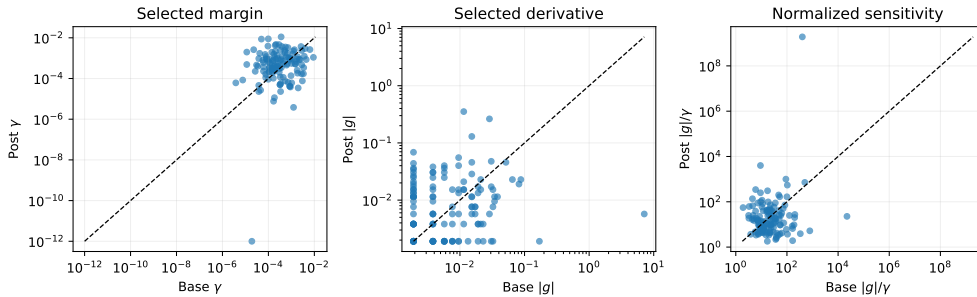


Figure 10: Nemotron Base–Post geometry at checkpoint-specific frozen target events. Unlike Qwen Thinking, none of the paired ratios differs significantly from one; nevertheless, local risk remains strongly predictive in both states.

1.26×10^{-7} to 3.35×10^{-8} , while next-token argmax remains unchanged. Routing transitions can therefore alter the full distribution without changing its top token.

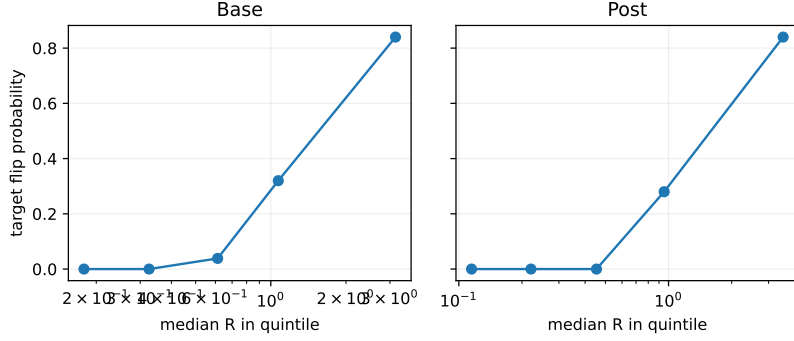


Figure 11: Nemotron exact target-flip rate over ascending within-model quintiles of frozen risk. Both Base and Post retain sharp risk ordering even though their checkpoint-level susceptibility contrast is unresolved.

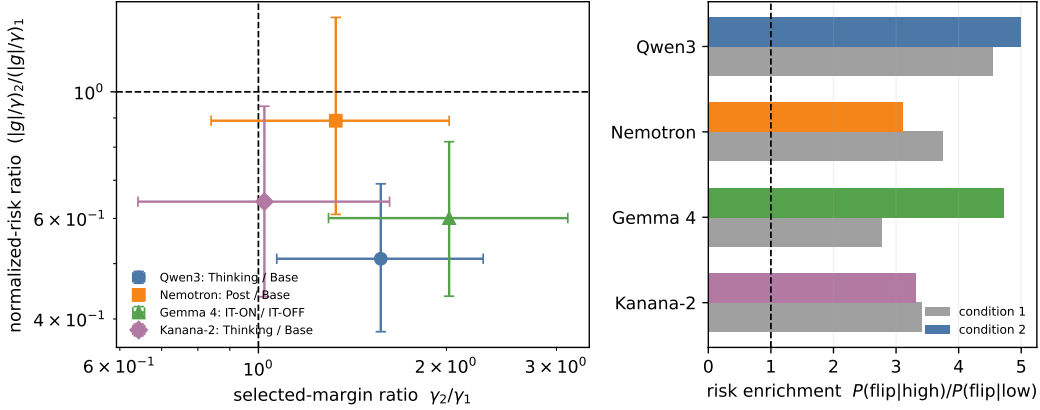


Figure 12: Risk detection is transferable while checkpoint/context variation is heterogeneous. Left: paired selected-margin and normalized-risk ratios for Qwen Thinking/Base, Nemotron Post/Base, fixed-weight Gemma IT-ON/OFF, and Kanana Thinking/Base; bars show 95% prefix- bootstrap intervals. Right: high/low risk enrichment remains above one before and after each comparison. Gemma changes rendered context, not weights.

6 Discussion

The experiments separate four levels that are often conflated. First, marginal source sensitivity is structured: $D_{\ell,i}$ has reproducible within-layer ranks and strong concentration. Second, source influence becomes relational only after specifying a context and a downstream boundary. The component ablation makes this precise: a transition requires the product of source susceptibility $|g|$ and boundary vulnerability γ^{-1} ; neither factor alone is an adequate targeting rule. Third, this relational structure is not globally stationary: aggregation loses edge stability, and checkpoints usually replace the competing boundary pair even at the same physical site. Fourth, the local field is behaviorally testable through retrospective localization, prospective targeting, and prospective ordering without max-selection.

This progression revises the initial geometry hypothesis without discarding its negative results. Full F is useful in a small routing-stable pilot, but it does not beat its diagonal and its absolute magnitude fails expanded-prefix robustness. The static graph similarly shows above-null signal but misses its scientific gate. The more appropriate abstraction is therefore not a single expert graph but a field $\mathcal{R}(x)$ whose source, destination, sign, and boundary identity depend on context and weights.

The routing clamp provides mechanistic triangulation. Baseline derivatives predict risky events; targeted interventions induce them; restoring expert selection removes most divergence among observed flip records. The first transition is usually sufficient, but nine exceptions warn against an additive cascade model. Together these results support the chain source perturbation $\rightarrow$ margin erosion $\rightarrow$ routing transition $\rightarrow$ expert replacement $\rightarrow$ output divergence.

The cross-model comparison separates measurement generality from field variation. The risk score remains prospectively predictive in all ten evaluated conditions, including Nemotron’s hybrid Mamba/attention/MoE and Gemma 4 MoE architectures. Its operating distribution, however, is checkpoint- and context-dependent. Ordinary Qwen A3B is statistically indistinguishable from Base. Qwen Thinking-2507 has lower absolute susceptibility but retains strong enrichment; its selected margins are larger, while broad derivative shrinkage remains unresolved. Nemotron Post retains risk enrichment but differs from Base on neither susceptibility nor the three geometry ratios. Gemma thinking-mode context changes geometry at fixed weights, primarily through margin expansion, without a resolved transition-incidence change. Kanana Thinking supplies the complementary derivative-associated case: selected margins remain stable while raw source-to-router sensitivity and normalized risk decrease significantly; its binary checkpoint contrast is directional but touches zero. Hence γ and $|g|$ are distinct variation channels rather than a single universal robustness axis. Checkpoint or mode changes can therefore reshape the field in two ways: they can replace the active boundary and, conditional on a selected competition, change its margin or derivative geometry. Our data do not support a universal direction for either channel. None of the checkpoint pairs is a randomized training treatment, and Gemma mode changes the rendered context.

7 Limitations

All primary interventions use GSM8K and source layer 23; layer and task transfer remain open. G4 establishes internal targeted routing transitions, not sequence correctness: next-token top-1 identity never changes in the cross-checkpoint grid, and no generated reasoning tokens are evaluated. High and matched-low sources are baseline-selected rather than randomized, so results demonstrate prospective targeting utility rather than the causal effect of assigning a risk score.

Weight-state comparisons are exploratory. Qwen Base, A3B, and Thinking-2507, the two Nemotron states, Gemma Base/IT, and Kanana Base/Thinking lack controlled training trajectories. Their differences are checkpoint associations, not causal effects of reasoning-oriented post-training. Expert IDs are not semantically aligned across checkpoints or families. Moreover, each condition independently selects its highest-risk target event. Geometry ratios therefore compare two risk-selected operating points, not an unconditional shift over all events. Our common-site audit freezes (ℓ, t) , but the actual rank-boundary expert pair almost always changes across weight states; it therefore aligns a physical router site rather than the same decision boundary. The 40.5 and 52.0 state-specific site-margin ratios reported in the appendix must not be interpreted as expansion of a shared boundary. Existing basis artifacts do not serialize nonselected event derivatives, so we explicitly avoid the stronger global-distribution claim. Gemma OFF/ON uses the same weights but different chat-template tokens and different prediction contexts; it is not an isolated causal thinking variable. Cross-family numerical levels are not directly comparable because routing top- k , source depth, shared- expert structure, tokenizer, architecture, and derivative oracle differ.

G2 and routing clamps reuse observed transition data and remain retrospective. Routing restoration is conditioned on flipping and is not an additive natural direct/indirect effect or population causal-effect estimate. The local quadratic pilot is small, while its 128-prefix expansion fails pooled absolute-magnitude robustness. Full-matrix, static-graph, and estimator redesign failures are retained rather than repaired post hoc. Finally, all findings are conditional on observed routing support, GSM8K prompts, and the four evaluated model families. The non-max experiment additionally candidate sampling on the 16 smallest-margin events per prefix; it supports prospective ordering in a near-boundary population, not global probability calibration over all eligible events.

Finally, these experiments intervene at the last content token of a GSM8K prompt and evaluate the immediate next-token distribution. They do not generate or intervene along reasoning trajectories. References to “Thinking” identify model or template conditions only; the present results establish prompt-boundary routing geometry, not a mechanism of chain-of-thought generation or reasoning accuracy.

8 Conclusion

MoE expert influence is structured but not globally stationary. Marginal source sensitivity is stable and concentrated, yet neither a full response matrix nor a population source-to-router graph yields the hoped-for robust global object. Signed local margin erosion instead supplies a predictive primitive whose two components are jointly necessary: neither large $|g|$ nor small γ alone targets transitions, while their normalized product does. Risk localizes old transitions, prospectively targets exact transitions on two disjoint splits, and orders frozen non-max candidates. It transfers across ten conditions in four MoE families and connects transitions to output divergence through routing restoration. Checkpoint and rendered-context differences are heterogeneous and act through both boundary identity and local geometry. These results motivate a view of sparse MoE computation as a context- and checkpoint-conditioned routing-risk field: measurement transfers, while the active competition and its operating point depend on weights and rendered context.

AI use statement

Generative AI tools assisted experimental code development, debugging, workflow documentation, analysis scripts, figure and table generation, and drafting and editing this manuscript. The authors reviewed AI-assisted outputs against saved configs and artifacts, executed numerical consistency checks, and retained negative results. No generative AI system is an author. The human authors take responsibility for the final content, claims, code, and artifacts. This disclosure must be updated before submission if subsequent uses change its scope.

Reproducibility statement

Supplementary code and artifacts specify checkpoint, pinned dataset revision, prompt and target-token protocol, split hashes, intervention definition, precision, seeds, and acceptance checks. One script regenerates every figure and result table from saved records. Appendices report derivative protocol lineage, negative gates, prospective selection hashes, routing-clamp validation, and estimator uncertainty. The reserved confirmation set is excluded.

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A Implementation validation

Table 8: Numerical and algebraic validation checks; all listed checks pass.

Check	Setting	Result
Zero intervention	layer 23, FP32	max logit difference 0
Inactive experts	layer 47, 20 cases	autograd and FD max 0
Finite difference	layer 47, 80 active	median error 0.735%
Finite difference	layer 47, 80 active	P90 error 4.942%
Finite difference	layer 47, 80 active	sign agreement 100%
Repeat score	fixed seed/prefix	max difference 0
F symmetry	$N = 512$	relative error 0
Streaming vs. offline	$N = 512$	1.57×10^{-16}
Positive semidefinite	$N = 512$	$\lambda_{\min} = -2.24 \times 10^{-20}$

Layer 47 was used for the formal finite-difference gate to minimize accumulated downstream roundoff; the same adapter and algebra are used at layer 23. An initial BF16 score path was rejected because its numerical discrepancy was too large; all reported response and intervention results use strict FP32.

B Detailed local prediction results

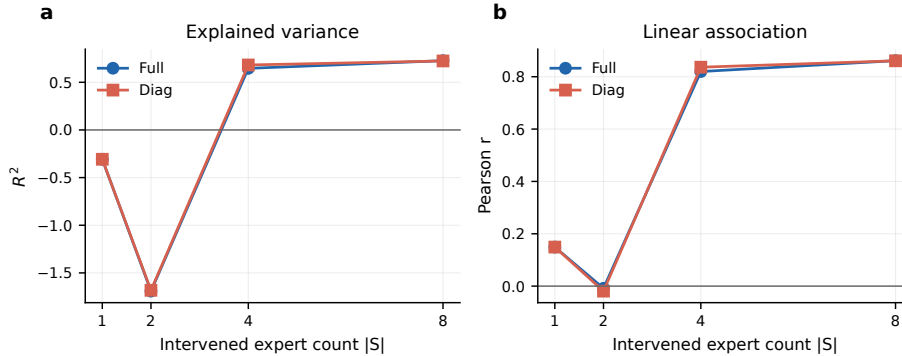


Figure 13: Descriptive local metrics by intervention subset size. Each stratum contains 10 interventions; the pooled result is primary.

Negative R^2 for small subsets reflects tiny observed KL and only 10 cases per stratum. Full and diagonal are identical for $|S| = 1$ by construction.

Table 9: Local prediction stratified by intervened expert count.

$ S $	Predictor	n	R^2	Pearson r	Spearman ρ_s
1	Full	10	-0.308	0.149	0.423
1	Diagonal	10	-0.308	0.149	0.423
2	Full	10	-1.691	-0.008	0.067
2	Diagonal	10	-1.682	-0.019	0.067
4	Full	10	0.646	0.820	0.503
4	Diagonal	10	0.682	0.836	0.503
8	Full	10	0.727	0.861	0.903
8	Diagonal	10	0.724	0.861	0.879

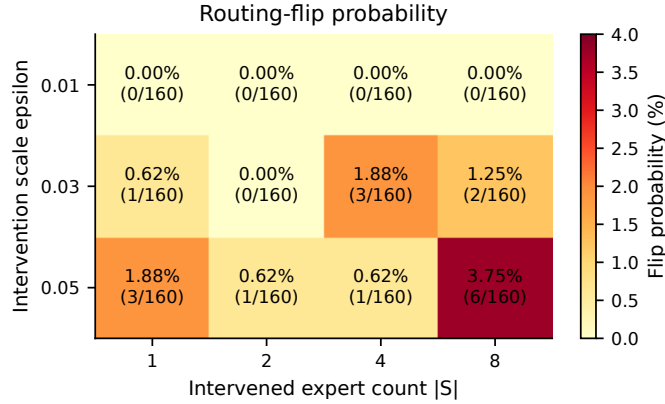


Figure 14: Flip probability by perturbation scale and subset size. Each cell contains 160 records. No record flips at $\epsilon = 0.01$. Percentages are descriptive because records share prefixes and interventions.

C Routing analysis details

D Sampling stability and estimator redesign

E Artifact scope

All figures are generated from saved JSONL, JSON, CSV, and NumPy artifacts. The pilot local prediction unit is an intervention mean over 16 prefixes; the E4 expanded unit is an intervention mean over 128 prefixes; the routing unit is one intervention-prefix record. These units remain separate to avoid inflating sample size. E4 artifacts are stored under artifacts/followup/e4_expanded_local_v2/ and include the 20,000- replicate bootstrap results. The generation manifest records output hashes and that the redesign confirmation set was not accessed.

F Derivative protocol lineage

The original stock sequence-wide G0 full-vector finite-difference gate failed. Component audits localized the discrepancy to the stock in-place expert aggregation: torch.func.jvp and low-level forward AD agreed with each other but not with reverse mode or stable finite differences. On the documented out-of-place index_add amendment, forward AD, low-level dual tensors, autograd.functional.jvp, reverse mode, and finite differences agree at the audited router event. Full-vector forward/reverse relative error is 1.67×10^{-6} with cosine effectively one; stable $\epsilon = 10^{-3}$ finite difference has relative error 0.0129 and cosine 0.99992.

Table 10: Record-level routing-transition summary.

Quantity	Estimate	Uncertainty/reference
Records	1,920	16 prefixes $\times$ 120 interventions
Records with flip	17	cluster rate CI [0.0021, 0.0203]
Affected interventions	14	of 120
ρ ROC-AUC	0.9977	Mann-Whitney $p = 4.24 \times 10^{-16}$
Median ρ , flip	33.415	no-flip P90 = 0.892
Mean KL, flip	6.822×10^{-4}	no-flip = 2.765×10^{-9}
Mean log-KL contrast	flip minus no-flip	cluster CI [6.77, 9.62]

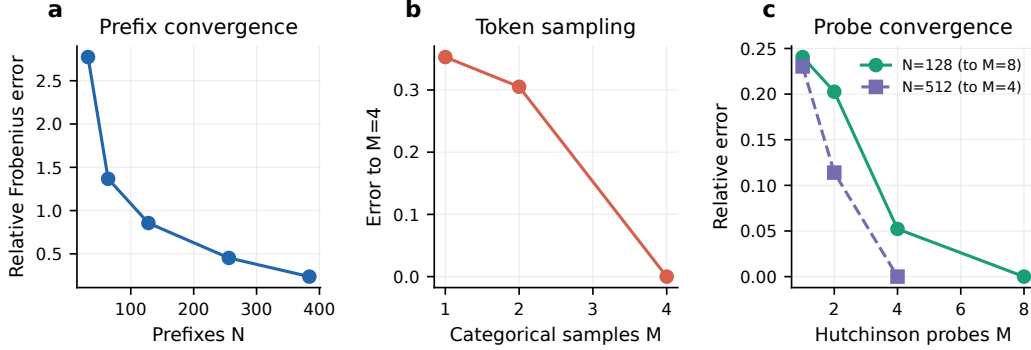


Figure 15: Sampling diagnostics: nested-prefix estimates, categorical token sampling, and Hutchinson probe convergence. Probe convergence does not imply convergence over independent prefixes.

A separate 8-prefix primal audit verifies that the amendment changes only expert aggregation, not parameters, routing policy, intervention definition, targets, or thresholds. Small FP32 differences are reported as accumulation-order effects rather than requiring bit identity. The stock failure remains recorded; it is not retroactively relabeled. The amended scalar router-margin derivative subsequently passes its primary AD oracle gate. Position-dependent full-vector FD remains an explicit numerical diagnostic, not a replacement gate.

G Event-level routing and clamp details

E5-A logs 2,206,080 downstream events for 120 interventions and 128 prefixes, forming 15,360 record aggregates. It finds 551 flip events in 225 records and 68 interventions. No flip satisfies $2\|\Delta q\|_\infty < \gamma$. Among affected records, 145/225 flips occur at the record’s minimum-margin event. Flip-event median margin is 0.003109 versus 0.091072 for nonflips; median router displacement is 0.03374. Flips concentrate in layers 45–46, followed by 42, 41, and 44. Event maxima reproduce legacy record-level ρ exactly.

First- and all-clamp KL are exactly equal in 216 records. The nine exceptions are retained. The minimum all-clamp suppression is -0.0106 ; one slightly negative case prevents an absolute restoration claim.

H Prospective selection and cross-architecture details

Full SHA256 values are stored in the released manifests. For compact audit, their leading 16 hexadecimal characters are: first G4 prefix/selection/outcome e45109de55c77ec, 287d825c913ed2ff, and be8aa423eee30525; second G4 prefix/selection/outcome 114a664f7a1131e4, 490d7ba687b3e948, and 650783637ac75daf. A3B selection/outcome begin bb615e3a3f6b8ab5/2478bcbdde1836e2; Thinking begins 1dc57f6e46fdb8f5/6377ee7825d3be1f. Corrected-score Nemotron Base selec-

Table 11: Prefix-split stability; lower difference and higher overlap are better.

Estimator	N	Half difference	Top-8 overlap	Decision
Categorical $M = 4$	512	0.762	0.613	unstable
Categorical RB $M = 4$	128	0.953	0.693	unstable
Hutchinson $M = 8$	128	0.892	0.564	unstable
Hutchinson $M = 4$	512	0.566	0.631	narrow gate fail

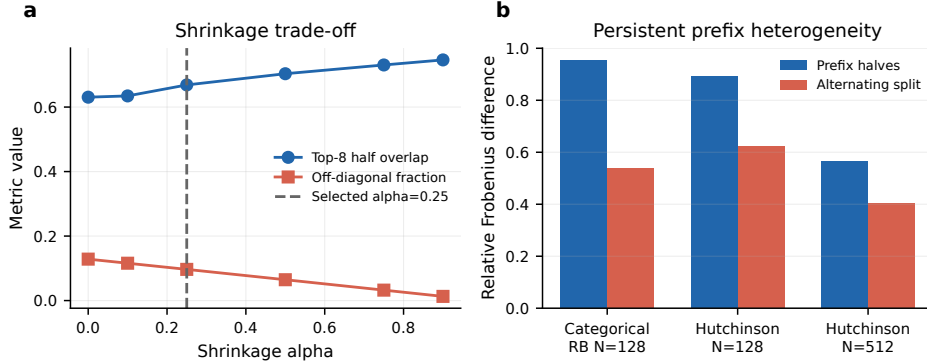


Figure 16: Exploratory redesign. Shrinkage trades subspace stability against off-diagonal magnitude, while prefix heterogeneity persists across estimators. Outcome tuning did not select a superior estimator, so confirmation stayed sealed.

tion/outcome begin 7b72886c99438242/ c3bfe808238a4ba9; Nemotron Post begins e1fc1b80a63d2626/fb89b604a1164e44. All baseline repeats, counts, target boundaries, finite values, and selection hash checks pass. No confirmation data are used.

P0 prospective freeze. The new P0 prefix, component-selection, component-outcome, non-max-selection, and non-max-outcome SHA256 values begin 58969e9184dc3819, f80c92e39385f701, 54b9ad4bff0117a5, 7d87e7a7763aff6d, and e0a44cef3471313f. The two 64-prefix subsets are disjoint from both earlier prospective G4 splits. Both outcome programs verify the frozen selection hash before loading the model and never recompute selection features.

For 128 paired Base-Thinking prefixes, selected-event γ is larger for Thinking in 73 cases (57.0%) and smaller in 55; the paired median ratio is 1.44. The median derivative over all eight active sources is smaller in 72 cases (56.2%) and larger in 56; its paired median ratio is 0.83. Hence the aggregate margin shift is statistically resolved but heterogeneous, not pointwise dominance. The stored basis contains all eight source directions only at the selected target event; it cannot recover the unconditional distribution over every nonselected eligible event.

Common-site boundary turnover. Freezing the Base-selected physical (ℓ, t) does not freeze the underlying decision boundary. Qwen Thinking preserves the Base rank-boundary expert pair in only 1/128 prefixes, and Kanana Thinking in only 2/128. At those physical sites, the state-specific Thinking/Base geometric-mean ratios are 40.5 [27.4,60.2] and 52.0 [35.9,75.4] for γ , 0.34 [0.28,0.41] and 0.29 [0.23,0.38] for maximum active-source $|g|$, and 0.0083 [0.0053,0.0129] and 0.0056 [0.0036,0.0087] for maximum $|g|/\gamma$. Because the boundary expert pair changes in nearly every prefix, these are state-specific physical-site operating-point contrasts, not same-boundary geometric effects.

Nemotron amendment and superseded path. Nemotron routes by $r_i = \sigma(q_i) + b_i$, not by raw logits. We therefore define its rank-6/rank-7 margin in corrected score space. An initial raw-logit-boundary analysis is retained only as a superseded diagnostic and is not reported as

Table 12: Exploratory no-flip shrinkage tuning smoke ($n = 11$).

Predictor	R^2	MSE	MAE
Shrinkage ($\alpha = 0.25$)	0.5042	4.1769×10^{-19}	4.5973×10^{-10}
Raw response	0.5057	4.1644×10^{-19}	4.5792×10^{-10}
Diagonal	0.4991	4.2201×10^{-19}	4.6515×10^{-10}

Table 13: Routing restoration over all 225 observed transition records.

Clamp	Median suppression	P10	Fraction > 0.5	Fraction > 0.9
First changed event	0.99677	0.85672	0.9600	0.8756
All changed IDs	0.99677	0.86761	0.9689	0.8844
All IDs and weights	0.99661	–	–	–

a scientific result. The corrected protocol independently freezes all 128 target events before modified forwards, perturbs only routed experts (not the shared expert), and uses a central finite-difference derivative with step 10^{-3} . Both corrected 384-record outcome grids pass selection-hash, uniqueness, finite-value, exact final-logit repeat, and exact target-boundary checks. Maximum absolute selection-to-outcome margin drift is 3.81×10^{-6} for Base and 8.39×10^{-5} for Post, while boundary expert identities remain exact.

Nemotron high-risk flip rates over ascending within-model $\mathcal{R}$ quintiles are $[0, 0, 0.038, 0.32, 0.84]$ for Base and $[0, 0, 0, 0.28, 0.84]$ for Post. These bins are diagnostic within-model calibration, not a shared probability scale. Their sharp ordering complements Table 6: the risk primitive transfers even though the Post/Base effect and geometry contrasts are unresolved. A multi-step FD audit at eight frozen events fails for both Nemotron states: relative to 10^{-3} , median relative L2 errors are 1.00 at 3×10^{-4} and 0.66–0.71 at 3×10^{-3} , with low sign agreement. The discrete derivatives are FP32 cancellation-limited. This negative numerical result does not alter the already frozen 10^{-3} prospective enrichment, but it rules out strong cross-step claims about Nemotron’s derivative magnitude or checkpoint shift.

Kanana architecture and derivative-associated variation. Kanana-2 uses a DeepSeek-V3-style MLA/MoE implementation with 48 decoder layers: layer 0 is dense and layers 1–47 are MoE. Each MoE layer has 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. We use physical source layer 23 and destinations 24–47, intervene only on routed contributions, and leave the shared MLP untouched. Expert identities are selected by $\sigma(q) + b$; contribution weights instead gather uncorrected sigmoid scores, renormalize over top-6, and multiply by 2.448. This explicitly separates ranking score from contribution weight. Stock/out-of-place aggregation relative L2 is 6.18×10^{-8} for Base and 6.04×10^{-8} for Thinking.

Base and Thinking selection/outcome hashes begin e3d6e497867da070/df86a9ca7703c59e and 3951599cfa5elf25/6da1b7ab6dc96a71. Each checkpoint uses a distinct 512-prefix atlas and a pre-outcome frozen selection. Both 384-record grids pass exact inactive/zero-delta, baseline repeat, boundary, finite-value, uniqueness, and selection-hash checks, with no relaxed matches. Ascending-risk flip rates are $[0, .038, 1, 1, .92]$ and $[0, 0, .231, .92, .96]$. The nonmonotonic top-bin fluctuation is compatible with 25–26 prefixes per bin; the defensible claim is sharp low-to-high stratification, not deterministic calibration.

Both checkpoints use the identical plain prompt rather than the Thinking chat template. Consequently the comparison is cleaner with respect to input context, but remains an observational weight-state association and does not evaluate generated reasoning. A 12-event follow-up varies the central-FD step over 3×10^{-4} , 10^{-3} , and 3×10^{-3} . Relative to 10^{-3} , median relative L2 errors are 0.055/0.010 for Base and 0.021/0.008 for Thinking; sign agreement and active-source Spearman are 1.0 throughout. The derivative stability gate therefore passes, closing the step-size concern while leaving the selected-event comparison limitation intact.

Table 14: Prospective non-max ordering conditional on near-boundary candidates. One downstream event is sampled per prefix before outcomes; all eight active source directions are retained. Intervals are prefix-cluster bootstrap 95% CIs.

Risk quintile	n	Frozen $\mathcal{R}$ range	Exact target flip
Q1	103	$[2.0 \times 10^{-6}, 7.8 \times 10^{-4}]$	.000 [.000,.000]
Q2	103	$[8.2 \times 10^{-4}, 2.9 \times 10^{-3}]$	.000 [.000,.000]
Q3	102	$[2.9 \times 10^{-3}, 8.1 \times 10^{-3}]$	.000 [.000,.000]
Q4	102	$[8.1 \times 10^{-3}, 4.0 \times 10^{-2}]$	.000 [.000,.000]
Q5	102	$[4.0 \times 10^{-2}, 3.20]$	.118 [.024,.219]

Table 15: Cross-checkpoint selected-event geometry and local susceptibility.

Checkpoint	Median γ	Median $ g $	Median $ g /\gamma$	Random flip	High/low
Base	0.00229	0.0787	30.44	0.1172	4.54×
A3B	0.00174	0.0664	29.00	0.1250	4.31×
Thinking-2507	0.00305	0.0547	15.91	0.0703	5.00×

Gemma architecture and mode-conditioned context. The executable Gemma 4 26B-A4B model has 30 MoE text layers, 128 routed experts, top-8 selection, and no shared expert. Its router first computes FP32 softmax probabilities, selects top-8 identities, then renormalizes selected weights and applies per-expert scales. We therefore define γ as the rank-8/rank-9 gap in the pre-selection softmax probabilities. Stock in-place and derivative-safe out-of-place aggregation differ by relative L2 7.09×10^{-8} . All three 512-prefix atlases and 384-record outcome grids pass inactive, zero-delta, repeat, boundary, finite-value, and selection-hash checks; no matched-low selection is relaxed.

Eight-event multi-step FD audits also pass for Base, IT-OFF, and IT-ON. Relative to the frozen 10^{-3} step, median relative L2 errors at 3×10^{-4} are 0.115, 0.102, and 0.070, and at 3×10^{-3} are 0.028, 0.034, and 0.024; sign agreement is 1.0 throughout and median active-source Spearman is at least 0.964. Thus all reported Gemma conditions pass the same numerical stability gate.

Table 16: Multi-step derivative audit. Each pair reports the median at $h = 3 \times 10^{-4}$ / 3×10^{-3} relative to the frozen 10^{-3} estimate. Spearman and sign agreement are over active source directions.

Condition	Relative L_2	Sign agreement	Spearman	Decision
Gemma Base	.115/.028	1.00/1.00	.976/.976	Pass
Gemma IT-OFF	.102/.034	1.00/1.00	.964/1.00	Pass
Gemma IT-ON	.070/.024	1.00/1.00	1.00/1.00	Pass
Kanana Base	.055/.010	1.00/1.00	1.00/1.00	Pass
Kanana Thinking	.021/.008	1.00/1.00	1.00/1.00	Pass
Nemotron Base	1.00/.706	.125/.500	.632/.696	Fail
Nemotron Post	1.00/.659	.167/.583	.514/.822	Fail

Gemma Base, IT-OFF, and IT-ON high-risk flip rates over ascending $\mathcal{R}$ quintiles are $[0, 0, 0, 0, 0.76]$, $[0, 0, 0.115, 0.92, 0.96]$, and $[0, 0, 0, 0.36, 0.96]$. OFF and ON use identical IT weights but distinct official chat renderings. In an audited prefix they contain 68 and 71 tokens and share no terminal suffix: OFF appends an empty thought block, whereas ON prepends a system turn with $\langle \text{think} \rangle$. Accordingly, the resolved ON/OFF geometry shift is evidence for mode-conditioned context dependence, not an isolated causal effect of thinking.

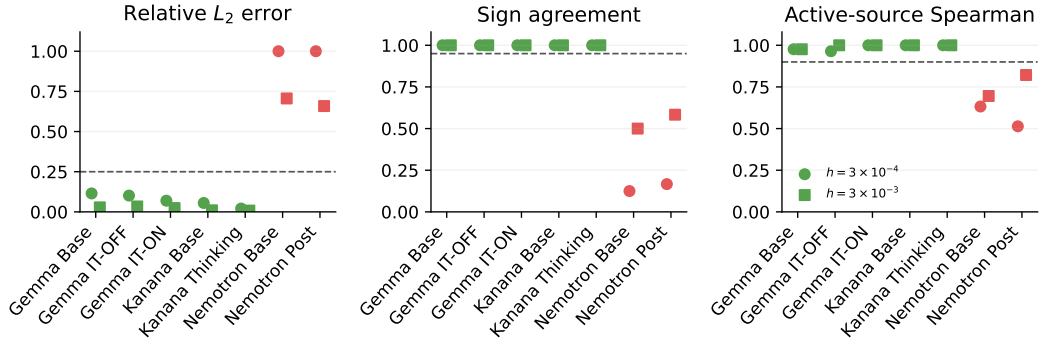


Figure 17: Multi-step finite-difference audit relative to the frozen $h = 10^{-3}$ oracle. Gemma and Kanana conditions pass the preregistered relative error, sign, and rank gates at 3×10^{-4} and 3×10^{-3} ; both Nemotron states fail. Green/red indicate the overall condition-level decision; dashed lines are acceptance thresholds.